LAB 3: PL/SQL part 2.

1. *Write a program in PL/SQL to calculate a period of employment of each employees from hr.employees. Assume, that all employees still work. Suggestion: to calculate the difference between two dates you can use subtraction operator; the result is expressed in days.*

```
BEGIN
  FOR emp_rec IN (
    SELECT employee_id, first_name, last_name, hire_date
    FROM employees
  ) LOOP
    DBMS_OUTPUT.PUT_LINE(
      emp_rec.employee_id || ' ' ||
      emp_rec.first_name || ' ' ||
      emp_rec.last_name || ' worked for ' ||
      ROUND(SYSDATE - emp_rec.hire_date) ||
      ' days'
    );
  END LOOP;
END;
/
```

The result:

```
100 Steven King worked for 8400 days
101 Neena Kochhar worked for 7573 days
102 Lex De Haan worked for 9285 days
103 Alexander Hunold worked for 7469 days
```

2. *Write a program in PL/SQL to find the number of rows effected by the use of update hr.employees set salary = salary * 1,5 where department = 50; Important: rollback statement must be executed in both the end of execution block and inside exception block!*

```
DECLARE
  v_rows NUMBER;
BEGIN
  UPDATE employees
  SET salary = salary * 1.5
  WHERE department = 50;
  v_rows := SQL%ROWCOUNT;
  DBMS_OUTPUT.PUT_LINE('Rows affected: ' || v_rows);
  ROLLBACK;
EXCEPTION
  WHEN OTHERS THEN
    ROLLBACK;
END;
/
```

```
Rows affected: 1
```

3. *Write a program in PL/SQL to calculate a period of employment of each employees from tables hr.employees. In this version you must take into account that table hr.job_history may also contain information about employment.*

Create job_history first:

```
CREATE TABLE job_history (
  employee_id NUMBER,
  start_date DATE,
  end_date DATE
);
```

Then some example data:

```
INSERT INTO job_history VALUES (101, DATE '2001-01-01', DATE '2003-01-01');
INSERT INTO job_history VALUES (103, DATE '2002-01-01', DATE '2004-06-01');
COMMIT;
```

```
BEGIN
  FOR emp_rec IN (
    SELECT employee_id, first_name, last_name, hire_date
    FROM employees
  ) LOOP

    DECLARE
      v_history_days NUMBER;
      v_total_days NUMBER;
    BEGIN
      SELECT NVL(SUM(end_date - start_date), 0)
      INTO v_history_days
      FROM job_history
      WHERE employee_id = emp_rec.employee_id;

      v_total_days :=
        (SYSDATE - emp_rec.hire_date) + v_history_days;

      DBMS_OUTPUT.PUT_LINE(
        emp_rec.employee_id || ' ' ||
        emp_rec.first_name || ' ' ||
        emp_rec.last_name || ' total employment: ' ||
        ROUND(v_total_days) || ' days'
      );
    END;

  END LOOP;
END;
/
```

The result:
100 Steven King total employment: 8400 days
101 Neena Kochhar total employment: 8303 days
102 Lex De Haan total employment: 9285 days
103 Alexander Hunold total employment: 8351 days

4. *Write a program in PL/SQL to calculate how many persons work (directly and indirectly) under a manager. The result should be presented in form of table:*
   a. Manager ID,
   b. Manager_First_name managers_last_name,
   c. Number of persons.

```
BEGIN
  DBMS_OUTPUT.PUT_LINE('Manager ID | Manager Name | Number of persons');

  FOR mgr_rec IN (
    SELECT employee_id, first_name, last_name
    FROM employees
    WHERE employee_id IN (
      SELECT DISTINCT manager_id
      FROM employees
      WHERE manager_id IS NOT NULL
    )
    ORDER BY employee_id
  ) LOOP

    DECLARE
      v_count NUMBER;
    BEGIN
      SELECT COUNT(*)
      INTO v_count
      FROM employees
      START WITH manager_id = mgr_rec.employee_id
      CONNECT BY PRIOR employee_id = manager_id;

      DBMS_OUTPUT.PUT_LINE(
        mgr_rec.employee_id || ' | ' ||
        mgr_rec.first_name || ' ' || mgr_rec.last_name || ' | ' ||
        v_count
      );
    END;

  END LOOP;

  ROLLBACK;

EXCEPTION
  WHEN OTHERS THEN
    ROLLBACK;
    DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
END;
/
```

The result:

```
Manager ID | Manager Name | Number of persons
100 | Steven King | 3
102 | Lex De Haan | 1
```